

Data Cleaning Project

Introduction

Before performing Exploratory Data Analysis, the data frame was carefully prepared to ensure accuracy and reliability. This process included changing data types, correcting errors, handling missing values, removing incomplete records, sorting out-of-sequence values, and resolving inconsistencies identified during the initial review. Additionally, a new column was created to enhance the depth and quality of the analysis. A properly preprocessed and well-structured data frame ensures that the subsequent analysis is robust, and that the conclusions drawn are meaningful and actionable.

2.1 Data Type Analysis and Conversion

To ensure consistency and accuracy, variables such as **age, quantity, earning per month, per-unit price, and total purchase amount** were converted from **float to integer** after handling missing (NaN) values. This order was necessary to prevent conversion errors and data loss. Since these values are represented as whole numbers, using the appropriate data types improves data integrity and supports more accurate analysis.

2.2 Deletion of Data

The following data were discarded to enhance the data frame's quality:

- Columns with no meaningful data, specifically **Tax and Unnamed**, were removed.
- **Four completely null rows** were discarded to prevent distortion in the analysis.
- **Duplicate rows** were also deleted to ensure that each record represented a unique transaction.

2.3 Addition of Data

The **Marketing Channel column** was added to capture how each customer discovered the product, enabling a direct comparison of lead sources and their contribution to sales. It includes the following channels: Email, Social Media, Referral, Ads, and Walk-ins.

2.4 Dataset Dimensions Before and After

- Original CSV file: **82 rows, 16 columns**
- Preprocessed CSV file: **76 rows, 15 columns**

2.5 Sorting Customer IDs

The Customer ID sequence contained an **out-of-place entry (CUST072)** that disrupted the expected order. To ensure proper **sorting**, rows were first filtered to retain only valid IDs matching the pattern CUST followed by digits, removing any summary or empty rows that could disrupt the sequence. This step was beneficial to improve clarity, ensure consistent ordering for visualizations, and prevent errors in cumulative calculations or sequence-dependent tasks.

2.6 Handling Missing Values

The following methods were applied to handle missing values in the dataset. The selection of each strategy was based on the data type of the variable and the proportion of missing values.

- For columns such as **education, feedback, mode of purchase, and province**, which are categorical (object type), missing values were filled using the **mode**, as it represents the most frequent category and preserves the typical characteristics of the data without introducing a meaningless category like 'Unknown'.
- The **gender column** was completed by inferring gender from the **customer name** column, using the map command, allowing for accurate and context-based filling.
- Missing values in **age and earning per month**, which are numerical, were filled using the **median**. The median was chosen because it is less sensitive to outliers and provides a strong estimate of central tendency for skewed distributions with a small number of missing values.
- The **quantity column** was completed by **dividing total purchase amount by per unit price**, ensuring consistency between related numerical fields.

2.7 File Saving

The cleaned file was saved in csv format as 'Project_clean_file.csv' for further visualization and analysis.